

Network Layer

This layer is responsible for logical addressing, routing & end-to-end packet delivery across interconnected networks.

- Handles logical addressing of devices
- Determines optimal routing path between networks
- Enables host-to-host communication across multiple networks.

IP Address:- unique numerical label assigned to each device connected to a computer network that uses Internet Protocol for communication. Its purpose:-

- Identifying a device on the network
- Locating the device to enable communication with other devices over a network like the internet.

Packet :- A packet is a structured block of data that includes both the information being sent & the information needed to deliver it across a network.

If data too large, & as they are split into many packets so sometimes packets are also a result of splitting (most of the time) on top of its main use functionality.

Structure of packet :-

- IP header :- contains control information used by networking devices and the destination host

Some of those fields are:-

- Source ip address
- Destination ip address
- version (IPv4, IPv6)
- Protocol
- etc

- Payload :- Contains the actual data being carried, which is often a TCP Segment or UDP datagram.

- TCP segment = TCP header + actual data

- TCP header = contains source port, destination port, seq, flags, etc.

→ same for UDP datagram (Header + data)

[Network devices like routers use packets, look at the source & destination ip addresses, find the best route & forwards the packets leading to either next routers or the destination host.]

(2)

Router :- Networking device that forwards data packets between different computer networks.

They also use broadcasting to gain the Layer 2 information (MAC address) before sending the packets and eventually forwarding it to switch of the different network.

Routing :- Process of selecting the ^abest path for data to travel across one or more networks.

Routing tables :- A table used by a router that contains routes to different networks.

Types of routing :-

Static routing :- manually configuring routes in routing tables

ex:- configuring network info like 192.168.0.0/24 of another network with the int. interface (Eth0, Eth1 or ip of different router's interface)

Dynamic routing :- everything in static but happens dynamically

Other Layers of OSI Model (Low Priority For Now)

Physical Layer :- Involvement of raw bits (0s, 1s)
physical mediums like ethernet cable, fibre
optic cables, wifi

Its job :- "How do I physically get these bits
from here to there?"

[Move bits through cables, wifi]

Data Link Layer :- delivers data between
devices on the same local area network.

- Involvement of MAC addresses
- This layer sends data as a frame
 - frame contains :- source MAC, destination
MAC, data

- Involvement of Switch (looks at MAC and
forwards frames)

- Address Resolution Protocol :- finds MAC
when you know the IP address

↓
[Broadcasting of "who has this IP" in
the network and then the reply packet
with "That's me, here's my MAC"]

(3)

Session Layer :- Handles session between two applications.

Presentation Layer :- Makes sure data is in a usable format.

[Its job $\rightarrow$ "How should this data be represented so the other side can understand it?"]

↓
uses encoding (ex:- text encoding for word "Hello" to bytes using UTF-8)

- Deals with encryption
- Deals with compression of data

OSI Model Order

1. Physical
2. Data Link
3. Network
4. Transport
5. Session
6. Presentation
7. Application

TCP/IP model

Both OSI and TCP/IP are conceptual things
the difference is TCP/IP is the actual networking
model and protocol suite used on the internet

TCP/IP Layers



Application~

Transport

Internet

Network Access

How They Map

OSI

Application

Presentation

Session

Transport

Network

Data Link

Physical

TCP/IP

Application~

Application~

Application

Transport

Internet

Network Access

Network Access

OSI view

- Application → Browser creates HTTP request.
- Presentation → Encryption (TLS), encoding.
- Session → Maintains the communication session.
- Transport → TCP establishes connection.
- Network → IP routes packets.
- Data Link → Ethernet/Wi-Fi frame.
- Physical → Electrical/radio signals.

TCP/IP view

- Application → HTTP + TLS + DNS.
- Transport → TCP.
- Internet → IP.
- Network Access → Ethernet/Wi-Fi.

Notice that TCP/IP combines several OSI layers.